Functional Forecasting of Turkish Electricity Load Curves

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Thesis Proposal

June 4, 2026

The Daily Load Curve: A Portrait of Economic Life

Why the curve has the shape it has:

- **Hours 0–5 (trough):** Most people asleep. Base-load plants (nuclear, hydro) dominate. Demand is low and very predictable.
- **Hours 6–8 (morning ramp):** Simultaneous awakening. Factories start shifts, HVAC switches on. Demand nearly **doubles in two hours**. Timing is sensitive to temperature and day type.
- **Hours 9–16 (plateau):** Industrial and commercial activity at full intensity. Predictable from calendar.
- **Hours 17–20 (evening surge):** People come home, cook, use AC or heating. Official forecast **most wrong here** (bias: 614–658 MWh/hour).
- **Hours 21–23 (wind-down):** Declining toward the trough.

The forecasting object

$Y_d(h) =$ Turkish electricity load on day d, hour $h = 0, \ldots, 23$

Not 24 separate quantities — a single daily **curve**.

Ramadan effect

During Ramadan, the entire shape shifts: iftar creates a massive evening spike, overnight demand rises sharply. The curve changes *regime*, not just level.

Who Uses These Forecasts and Why Accuracy Matters

The Turkish electricity market (post-2013 liberalisation):

- **TEİAŞ** (TSO): publishes the official day-ahead load forecast used to schedule grid operations and clear the day-ahead market (EPIAS). *This is our benchmark.*
- **Generators** bid into the day-ahead market based on expected demand. Better demand forecasts ⇒ more strategic bids.
- **Large industrial consumers** face real-time imbalance penalties when their consumption deviates from contracted amounts. Better forecasts reduce penalties.
- **Unit commitment:** TEİAŞ decides which power plants to start up (hours in advance, significant startup costs). A better forecast ⇒ fewer unnecessary startups.

The systematic underprediction puzzle

Year	Off. bias	Our bias
2023	+344 MWh	+176 MWh
2024	+585 MWh	+34 MWh
2025	+472 MWh	+98 MWh
Avg.	+467 MWh	+103 MWh

The official forecast **always underestimates** actual demand. The gap worsened in 2024 (hot summer + demand growth). Our rolling-origin model adapts continuously and nearly eliminates the bias in 2024 (+34 MWh).

Why does the official underpredict? Likely: model staleness, growing demand, asymmetric institutional incentives (over-procurement is visible; real-time reserve costs are not).

From 24 Numbers to a Curve: Why FDA?

The standard approach: fit 24 separate forecasting models, one per hour. Problems:

- Ignores the shape of the curve — the morning ramp is a *joint* feature of hours 6, 7, and 8, not an independent event.
- Cannot borrow information across hours (hour 6 today predicts hour 7 today, not just hour 6 tomorrow).
- Cannot easily model "shape features" like ramp steepness or peak timing.

The functional approach: treat each day as a single mathematical object — a curve $Y_d(\cdot)$.

- Preserves intraday shape information.
- Allows differentiation: $DY_d(h) = \frac{d}{dh} Y_d(h)$ measures how fast demand is *changing* at each hour.
- FPCA compresses 24 correlated observations to $K = 2$ uncorrelated scores that carry 97.3%

Representation: B-spline basis ($M = 12$ functions)

$$Y_d(h) \approx \sum_{m=1}^{12} c_{dm} B_m(h)$$

then FPCA:

$$Y_d(h) = \mu(h) + \sum_{k=1}^{K} \xi_{dk}\, \phi_k(h) + \varepsilon_d(h)$$

Forecast reconstruction

Once we forecast the K scores $\hat{\xi}_d$, the full 24-hour load curve is recovered as

$$\hat{Y}_d(h) = \hat{\mu}(h) + \hat{\Phi}(h)' \hat{\xi}_d$$

FPCA: Compressing the Curve to Two Numbers

Variance decomposition (full training data):

FPC	Load var.	Cum.	Meaning
1	93.4%	93.4%	Level
2	3.9%	97.3%	Shape
3	1.1%	98.4%	—
4	1.0%	99.4%	—

$K = 2$ exceeds the 95% cumulative variance rule while keeping the score vector parsimonious.

Stability check (script 15): We fit FPCA on 2019–2021, 2020–2022, and 2022–2024 separately. All pairwise inner products > 0.993. *The eigenfunctions do not change* — static FPCA is empirically justified.

Economic meaning of the two components:

FPC 1 ("Level", 93.4%): Its eigenfunction is nearly flat across all 24 hours. A high score ξ_{d1} means uniformly high consumption — a busy industrial day, cold winter, hot summer. *Think of it as: how much electricity was used today overall.*

FPC 2 ("Shape", 3.9%): Its eigenfunction is positive at morning and evening peaks, negative at midday. A high ξ_{d2} means pronounced morning/evening peaks relative to midday — typical of residential-heavy days (weekends, holidays). A low ξ_{d2} means a flatter profile — typical of summer when AC smooths the curve. *Think of it as: how "peaky" today's curve is.*

The Score Forecasting Model

After FPCA, the forecasting problem is: *given the score history, predict* (ξ_{d1}, ξ_{d2}) *for tomorrow.*

FPCA VAR(1,7) — core score dynamics:

$$\xi_d = a + A_1\xi_{d-1} + A_7\xi_{d-7} + u_d$$

- **Lag 7:** Monday demand resembles last Monday's. Weekly seasonality is the dominant pattern in electricity use.
- **Lag 1:** yesterday's level and shape adjust the weekly baseline for recent demand shifts.
- Estimated by OLS equation by equation.

Augmented model:

$$\xi_d = a + A_1\xi_{d-1} + A_7\xi_{d-7} + Bz_d + u_d$$

Feature blocks in z_d (all predetermined at date d):

- **Calendar:** weekend, annual sine/cosine, summer break
- **Holiday:** fixed holidays, Ramadan, Bayram (Eid), holiday eves, post-holiday, bridge days
- **Load shape:** yesterday's mean, peak, range, rolling 7- and 28-day averages
- **Derivative features:** slope statistics from $DY_{d-1}(h) = \frac{d}{dh}Y_{d-1}(h)$ (mean slope, volatility, min/max, morning ramp, evening ramp, end-of-day drop)

Holiday and derivative features are the two most important additions. Without holiday indicators, the model does not beat the official forecast.

Evaluation Design: No Leakage, No Shortcuts

Rolling-origin evaluation (walk-forward cross-validation for time series):

1. Test period: Jan 1 2023 – Dec 31 2025 (1,096 days)
2. For each test day d:
   - Estimate FPCA on all data before d
   - Estimate VAR + features on all data before d
   - Produce one-day-ahead forecast for d
   - Record error
3. Total: 26,304 hourly forecast errors evaluated

At every step, **only information dated $< d$ is used**. This mimics real operational conditions.

Metrics:

$$\mathrm{MAE} = \frac{1}{N}\sum_{d,h}|e_{dh}|, \quad \mathrm{MAPE} = \frac{1}{N}\sum_{d,h}\frac{|e_{dh}|}{Y_d(h)}$$

- **MAE:** absolute error in MWh; directly interpretable
- **MAPE:** percentage error; scale-free; standard in the electricity forecasting literature
- **RMSE:** penalises large errors more

Statistical testing: Diebold-Mariano test (H_1: challenger has lower expected absolute error than official). Applied pooled (all 26,304 observations) and hour by hour (24 separate tests).

Comparing MAE numbers without a significance test is not enough.

Main Results

Model	MAE	MAPE	MAE gain	MAPE gain	DM
Seasonal naive Y_{d-7}	1912	5.13%	−57.6%	−62.9%	—
Official forecast	1214	3.15%	0%	0%	ref.
FPCA VAR(1,7) scores	1443	3.76%	−18.9%	−19.4%	×
+ ramp/calendar	1238	3.26%	−2.1%	−3.7%	×
+ holiday/load-shape	1001	2.63%	+17.5%	+16.5%	***
+ derivative features	**980**	**2.56%**	+19.2%	+18.7%	***
+ residual step	653	1.70%	+46.2%	+46.0%	***

DM: Diebold-Mariano test vs. official forecast (pooled hourly, H_1: challenger better). *** $p < 0.001$; $\times$ = not significant or worse than official.

- **Key insight:** the pure VAR dynamics are not enough. Holiday and load-shape features are what make the model competitive.
- **MAPE and MAE tell identical stories** — the gains are not a scale artifact.
- Main model beats official by **19% MAE** and **19% MAPE**, with $p \approx 10^{-163}$.

Where the Model Wins and Where It Loses

MAPE gain by hour (main model vs. official):

Hours	Regime	MAPE gain
0–5	Overnight trough	+14% to +60%
6–8	Morning ramp	−6% to −24%
9	Ramp end	$\approx$ 0%
10–16	Daytime plateau	+20% to +35%
17	Evening transition	+7%
18–19	Evening surge	−7% to −18%
20–23	Night wind-down	+10% to +20%

The ramp-hour puzzle

Our model **loses** at hours 6–8 (morning ramp) and 18–19 (evening surge) — the steepest transition hours.

Why? The official forecast incorporates weather (temperature) forecasts that determine the *timing* of ramps. Our model uses only historical load patterns.

A one-hour error in ramp timing propagates across the entire ramp — the slope error is not local.

DM test: model beats official at **19 of 24 hours**, statistically significant at **18 of 24 hours** (5% level). The residual step wins at **24 of 24 hours**.

Diagnostics: Why the Residual Step Is Motivated

ACF of forecast errors (Ljung-Box, lag 14):

Series	LB statistic	Conclusion
Hour 0 errors	3,124	Autocorrelated
Hour 6 errors	7,227	Strongly autocorrelated
Hour 8 errors	994	Autocorrelated
All 24 hours	—	All reject H_0

Interpretation: errors at ramp hours (especially hour 6) are correlated across consecutive days. If the model misses the ramp timing on Monday, it tends to miss it on Tuesday too. *This is a predictable pattern.*

The residual correction

For target day d, using only past data:

$$\hat{e}_s(h) = Y_s(h) - \hat{Y}_s^{base}(h), \quad s < d$$

Estimate a correction curve $\hat{r}_d(h)$ from recent residuals and apply:

$$\hat{Y}_d^{res}(h) = \hat{Y}_d^{base}(h) + \lambda_d \, \hat{r}_d(h)$$

λ_d is chosen by rolling-origin cross-validation on pre-d data only. **No leakage.**

Result: 46% MAE gain, 46% MAPE gain vs. official. Wins at all 24 of 24 hours including the 5 ramp hours where the base model lost.

Validation and Robustness

1. Diebold-Mariano test (pooled):

Model	DM p-value
FPCA VAR(1,7) only	≈ 1 (worse)
+ ramp/calendar	0.996 (worse)
+ holiday/load-shape	10^{-135}
+ derivative features	10^{-163}
+ residual step	≈ 0

The gains are not sampling variation.

2. Eigenfunction stability:

- Fit FPCA on 2019–21, 2020–22, 2022–24 separately.
- Pairwise inner products: all > 0.993.
- Static FPCA is empirically justified — no need for dynamic FPCA.

3. $K = 4$ robustness:

K	Resid.	MAE	MAPE	MAE gain
2	No	980	2.56%	19.2%
2	Yes	653	1.70%	46.2%
4	No	801	—	34.0%
4	Yes	**640**	—	**47.3%**

$K = 2$ is the conservative baseline (parsimony + variance rule). $K = 4$ is a robustness check.

4. Random forest benchmark:
Replace OLS with RF on the same feature vector z_d (same FPCA structure, same rolling-origin design). Tests whether the linear score equation misses nonlinearity. *[Result pending.]*

Thesis Contribution

The coherent story in one flow:

1. Turkish daily electricity load is a **curve**, not 24 separate numbers. FDA preserves intraday shape.
2. FPCA compresses to **2 scores** (97.3% variance). Eigenfunctions are empirically stable.
3. Scores follow a **VAR(1,7) + features**. Holiday and load-shape features are essential — pure dynamics fail.
4. The model beats the official forecast by **19% MAE / 19% MAPE**, $p \approx 10^{-163}$.
5. It loses at **ramp hours** (6–8, 18–19) where the official has a weather-forecast advantage.
6. **Error ACF** shows remaining serial structure. The residual step exploits this: 46% MAE gain, 24/24 hours.

Methodological contributions:

- Functional representation + FPCA score forecasting applied to the Turkish grid
- Systematic feature engineering from functional derivative
- Rigorous rolling-origin design with DM testing
- Empirical test of eigenfunction stability

Economic contributions:

- Document and explain the official forecast's systematic underprediction
- Identify ramp hours as the forecasting frontier (weather vs. historical patterns)
- Quantify the value of holiday and shape information for grid operators and market participants

Selected References

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